

S-BFP Artifact Appendix

From Frozen Pose to Live Dance: Stochastic Browser Fingerprinting for Robust Risk-Based Authentication

Artifact overview

S-BFP is a Flask and browser research prototype that turns a static browser fingerprint into a server-controlled challenge-response measurement. A deterministic random bit generator derives device-specific primitives, while a session input changes their spatial arrangement. The artifact implements Canvas text, Web Audio waveforms, and WebGL triangles.

The package supports (1) functional inspection of all three browser workflows and (2) independent recomputation of the paper's stability and environment tables from a de-identified 213-participant summary dataset.

Claims and evaluation mapping

Paper claim	Evaluation action	Expected result
Audio stability	Run reproduction script	206/206 (100.0%)
Canvas stability	Run reproduction script	196/198 (99.0%)
WebGL stability	Run reproduction script	177/193 (91.7%)
Cohort distribution	Inspect generated Table 3 CSV	213 devices; exact row match
Executable workflow	Run all three browser panels	Repeated render and stability summary

Scope

The artifact does not implement a complete production risk engine and does not claim to prove security against every adaptive attacker. WebGL instability is an evaluated result, not necessarily an execution failure. The restricted per-participant source collection is not part of the release package.

Requirements and installation

Hardware: any x86-64 or ARM64 CPU, 1 GiB free memory, and about 300 MiB disk for native use. A discrete GPU is not required. Software: Python 3.11-3.13 and a current browser with Canvas, Web Audio, and WebGL. Docker with Compose v2 is optional.

Windows PowerShell

```
.\scripts\setup.ps1  
.\scripts\run.ps1
```

Linux or macOS

```
sh scripts/setup.sh  
sh scripts/run.sh
```

Docker alternative

```
docker compose up --build
```

Open <http://127.0.0.1:5001/>. Native startup normally opens this address automatically. The aggregate reproduction is CPU-only and typically finishes in under one minute. Interactive rendering usually takes several seconds per modality.

Functional evaluation

1. Select Register and enter a pseudonymous username of 3-20 letters, digits, or underscores.
2. Read and affirm the consent notice, then save the generated password.
3. Acquire the exclusive test session.
4. Run Canvas, Audio, and WebGL. Canvas performs five repetitions; Audio and WebGL perform ten.
5. Confirm that every panel presents a completion message. A changed hash is a legitimate instability result.
6. With default settings, the new runtime record must not contain an IP address, full user-agent string, or raw Canvas fingerprint.

Paper-table reproduction

```
python scripts/reproduce_tables.py --verify-paper
```

Expected reproduction result

Expected values: Audio 206/206; Canvas 196/198; WebGL 177/193; environment total 213.

Automated and package checks

```
python -m unittest discover -s tests -v
python scripts/build_artifact.py
python scripts/verify_artifact.py dist/s-bfp-usenix-artifact.zip
```

The tests use an isolated temporary data directory. The package verifier requires the documentation and public dataset, verifies SHA-256 entries, checks the 213-row schema, and rejects restricted source, runtime-data, upload, cache, and Git paths.

Released data

data/public/records.jsonl is a deterministic aggregate of each participant's first collection session, matching the paper's cohort snapshot. Each row contains only a new sequential ID, coarse operating-system and browser categories, and modality availability, run count, unique-result count, and stability boolean.

Original usernames, password material, IP addresses, full user agents, timestamps, renderer strings, seeds, rendering hashes, raw images, and audio samples are omitted. Evaluators do not need the restricted source records.

Limitations and troubleshooting

WebGL depends on the evaluator's browser and graphics stack. If unavailable, enable browser acceleration or use a current Chromium/Firefox build; aggregate reproduction does not require WebGL. If port 5001 is occupied, set `S_BFP_PORT`. Docker reproduces the server environment, not the host browser renderer. The process-local session coordinator supports one active evaluator and is not a multi-worker production design.

Submission notes

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